Team Name: CSE4104-7E-T02

Section: 7E

Project Title: AI-Powered Prescription Recognition and Intelligent Medication Reminder System

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Preliminary Project Description:

The proposed system is an AI-powered web application that reads photographs of handwritten prescriptions and automatically generates medicine reminders. It extracts medicine names, doses and timing patterns such as "1+0+1", lets the user confirm the schedule, then sends timely reminders and tracks taken or missed doses. Users can also view general medicine information and prices, along with a cautious note to consult a doctor or pharmacist if a medicine does not seem to match their condition. Unlike existing tools that offer either handwriting recognition or reminders alone, this system combines both for Bangladeshi prescriptions. It aims to help ordinary people, especially elderly patients and caregivers, avoid missed doses and better understand their treatment.

Why AI is needed in this project:

Bangladeshi prescriptions are mostly handwritten, mix English and Bangla, and use shorthand dose patterns like “1+0+1”. Without AI, users must type every medicine and timing manually, which is slow and error-prone. A multimodal AI model reads the prescription photo and extracts medicine names, doses, durations and timing as structured data. The system then builds the reminder schedule automatically, and the user only confirms or edits it. AI also gives plain-language explanations of each medicine and gently flags a medicine that doesn’t match the stated condition, advising the user to consult a doctor or pharmacist. Rule-based code can’t handle this variation in handwriting, abbreviations and brand names.

Initial Technology Stack:

- **Frontend:** React (or Next.js) as a responsive PWA, styled with Tailwind CSS.
- **Backend:** Node.js with Express, JWT authentication with bcrypt, and node-cron or BullMQ for reminder scheduling.
- **Database:** PostgreSQL, with cloud storage (Supabase Storage or Cloudinary) for prescription images.
- **AI Tools:** Google Gemini API (multimodal) for prescription extraction, medicine summaries and the condition check, plus a seed database of common medicines for price and generic data.

Similar Existing Systems/Research:

Name	What it does	Shortcoming our project fills
memorAIs (Shaveet et al., 2023)	Reads medicine bottle labels with OCR and exports a calendar reminder file.	No accuracy results, doesn't extract medicine names, and the server is offline.
Smart Medicine Reminder System Using OCR with Family Monitoring (Tari et al., 2026)	A Flutter and Firebase app that scans prescriptions and sends reminders and family alerts.	No measured accuracy, and no medicine information or price.
Multi-Language Prescription Recognition (Pavithiran et al.)	A deep learning system that reads prescription images and matches them to a pharmacy database.	No accuracy for its own system, and no reminders.
Handwritten Medical Term Recognition (Scientific Reports, 2022)	Recognizes handwritten medical terms from Bangladeshi doctors (93% accuracy).	Needs a smartpen, not a phone photo, and gives no reminders.
Doctor's Handwritten Prescription BD dataset	4,680 word images of 78 medicines from Bangladeshi prescriptions.	A dataset only, covering a small set of medicines, and no application.
MIRAGE (Mankash et al., 2024)	Fine-tuned multimodal models extract medicines and doses from handwritten prescriptions.	Tested on simulated data, and the authors say it is not deployable. Your confirm-and-edit step covers this.
Hybrid OCR for Handwritten Prescriptions (Ameer et al., 2026)	An OCR pipeline for prescriptions, compared against GPT-4o.	Text extraction only, with no schedule, reminders or medicine information.
Gemini Prescription Parser (Sotherden, 2025)	A blog demo that turns prescription images into structured data with Gemini.	A backend demo with no user app, reminders or Bangladesh focus.
PharmaScan (Tiwari)	An Android demo that describes a photographed medicine using Gemini.	A proof of concept with no prescription reading or reminders.
Medisafe	Global reminder app with refill alerts, family sharing and US drug-interaction checks.	No prescription photo scanning, and not built for Bangladesh.
DigiCure BD	Bangladeshi app with reminders, stored prescriptions and appointments.	No scanning or medicine information.
Daktarbhai	Bangladeshi app with reminders, e-prescriptions and doctor consultations.	No AI reading of paper prescriptions.
MedEx Plus	Bangladeshi medicine index with details and price sorting.	No reminders or prescription scanning.
TrueMed	A concept where pharmacists explain prescriptions and set reminders.	Manual and slow, with no AI.